```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
```

```
uploaded = files.upload()

video_path = list(uploaded.keys())[0]

print("Uploaded video:", video_path)
```

Choose Files 11956377_... (1) (1).mp4
11956377_1080_1920_25fps (1) (1).mp4(video/mp4) - 24877105 bytes, last modified: 9/11/2026 - 100% done
Saving 11956377_1080_1920_25fps (1) (1).mp4 to 11956377_1080_1920_25fps (1) (1).mp4
Uploaded video: 11956377_1080_1920_25fps (1) (1).mp4

```
cap = cv2.VideoCapture(video_path)

ret, frame1 = cap.read()
ret, frame2 = cap.read()

if not ret:
    print("Error: Could not read video.")
else:
    print("Video loaded successfully!")

gray1 = cv2.cvtColor(frame1, cv2.COLOR_BGR2GRAY)
gray2 = cv2.cvtColor(frame2, cv2.COLOR_BGR2GRAY)

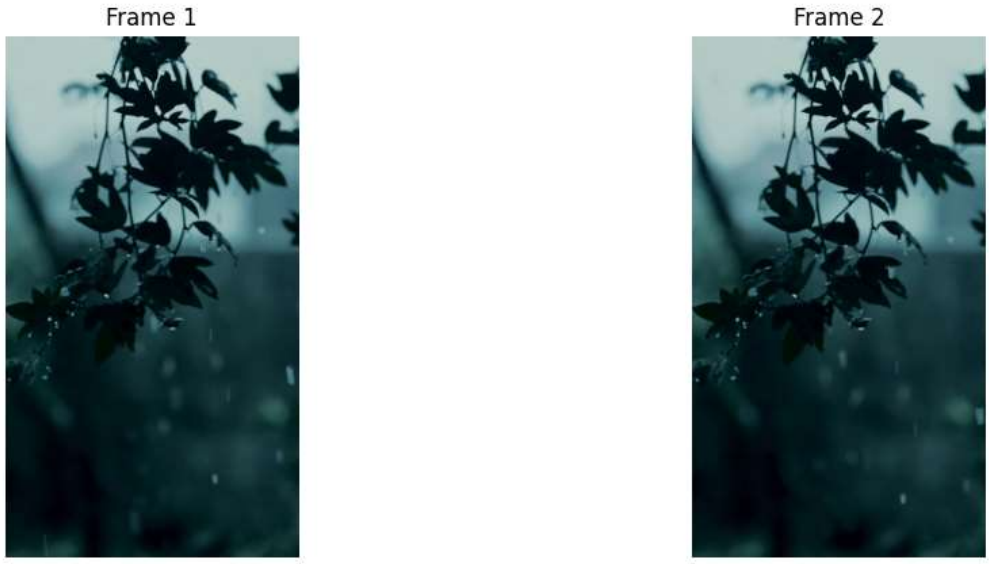
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(frame1, cv2.COLOR_BGR2RGB))
plt.title("Frame 1")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(cv2.cvtColor(frame2, cv2.COLOR_BGR2RGB))
plt.title("Frame 2")
plt.axis("off")

plt.show()
```

```
Video loaded successfully!
```




```
feature_params = dict(
    maxCorners=100,
    qualityLevel=0.3,
    minDistance=7,
    blockSize=7
```

```
)

p0 = cv2.goodFeaturesToTrack(gray1, mask=None, **feature_params)

print("Number of feature points detected:", len(p0))
```

```
Number of feature points detected: 40
```

```
lk_params = dict(
    winSize=(15, 15),
    maxLevel=2,
    criteria=(
        cv2.TERM_CRITERIA_EPS | cv2.TERM_CRITERIA_COUNT,
        10,
        0.03
    )
)

p1, st, err = cv2.calcOpticalFlowPyrLK(
    gray1,
    gray2,
    p0,
    None,
    **lk_params
)

good_new = p1[st == 1]
good_old = p0[st == 1]

print("Successfully tracked points:", len(good_new))
```

```
Successfully tracked points: 40
```

```
result_lk = frame2.copy()

for new, old in zip(good_new, good_old):

    x_new, y_new = new.ravel()
    x_old, y_old = old.ravel()

    cv2.arrowedLine(
        result_lk,
        (int(x_old), int(y_old)),
        (int(x_new), int(y_new)),
        (0, 255, 0),
        2,
        tipLength=0.3
    )

    cv2.circle(
        result_lk,
        (int(x_new), int(y_new)),
        4,
        (0, 0, 255),
        -1
    )

plt.figure(figsize=(12,7))
plt.imshow(cv2.cvtColor(result_lk, cv2.COLOR_BGR2RGB))
plt.title("Lucas-Kanade Sparse Optical Flow")
plt.axis("off")
plt.show()
```




```
flow = cv2.calcOpticalFlowFarneback(
    gray1,
    gray2,
    None,
    0.5,
    3,
    15,
    3,
    5,
    1.2,
    0
)

print("Farneback Dense Optical Flow calculated successfully!")
```

```
Farneback Dense Optical Flow calculated successfully!
```

```
magnitude, angle = cv2.cartToPolar(
    flow[..., 0],
    flow[..., 1]
)

hsv = np.zeros_like(frame1)
hsv[..., 1] = 255

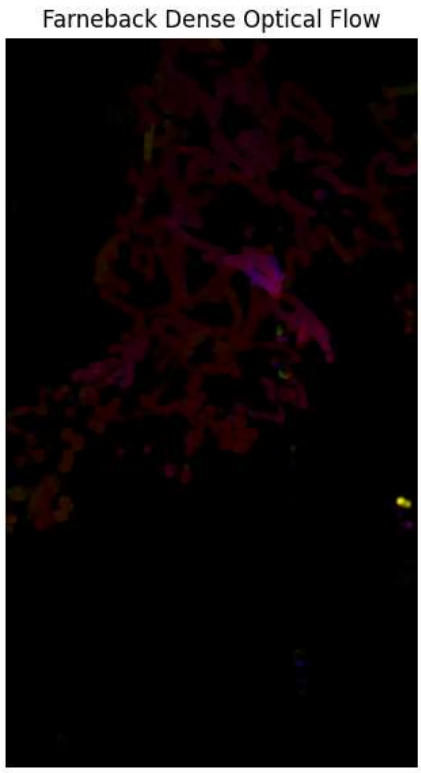
hsv[..., 0] = angle * 180 / np.pi / 2

hsv[..., 2] = cv2.normalize(
    magnitude,
    None,
    0,
    255,
    cv2.NORM_MINMAX
)

dense_flow = cv2.cvtColor(
    hsv,
    cv2.COLOR_HSV2RGB
)

plt.figure(figsize=(12,7))
plt.imshow(dense_flow)
plt.title("Farneback Dense Optical Flow")
```

```
plt.axis("off")
plt.show()
```




```
plt.figure(figsize=(15,6))

plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(result_lk, cv2.COLOR_BGR2RGB))
plt.title("Lucas-Kanade Sparse Optical Flow")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(dense_flow)
plt.title("Farneback Dense Optical Flow")
plt.axis("off")

plt.show()
```

